DM

Lecture - 1

→ **Discrete Maths**

→ **Continuous Vs Discrete**

→ DM is Branch of Maths which deals with objects that can consider distinct and separated values.

- Set theory
- functions, Relations
- Probability
- Combinations
- Graph and Trees

} DM

* Applications of DM

- Developing efficient Algo.
- Problem Solving Skills
- Cryptography
- Formulate research Problem

- Text Book: Elements of Discrete Mathematics by C. L. Liu

Ch - 1

Set and Propositions

→ "There are 4^{th} Sem students in this meeting of CE Branch"

→ Objects :- 4^{th} Sem Students
- Students
- CE Branch Students
- 4^{th} sem + CE + Students

→ Set : Way to organize objects (Discrete)

→ Set :- "Collection of Distinct objects"

→ $S1 = \{1, 2, 3, a, b\}$
$S2 = \{x \mid x \text{ is positive even No.}\}$

OR

$S2 = \{2, 4, 6 \ldots\}$

→ Empty Set : "Set without any element"

$\Rightarrow \{\}$, OR $\emptyset$

→ Cardinality of Set [| |]

No. of elements (items) in a Set

→ $S1 = \{1, 2, 3, 4\}$ $|S1| = 4$

→ $S2 = \{1, \underbrace{\{a, b\}}_{\text{set}}\}$ $|S2| = 2$

→ $S3 = \{\underline{\{\}}\}$ → $|S3| = \underline{1}$

* Belongs to ($\in$)

$S1 = \{John, mary\}$

$S2 = \{\{John, mary\}\}$

$S3 = \{\{\{John, mary\}\}\}$

- $John \in S1$ – ✓
- $John \in S2$ – ✗
- $John \in S3$ – ✗
- $S1 \in S2$ – ✓
- $S2 \in S3$ – ✓
- $S1 \in S3$ – ✗

* Subset : ($\subseteq$)

P is Subset of Q if every element of P is also element of Q

$P = \{1, 2\}$ $Q = \{1, 2\}$

$\Rightarrow P \subseteq Q$

* Proper Subset [$\subset$]

→ $P \subset Q$ if P is Subset of Q and $P \neq Q$.

$P = \{1, 2\}$ $Q = \{1, 2, 3, 4\}$

$P \subset Q$

* Set Properties

1. For any Set P, $P \subseteq P$

2. Empty Set is Subset of every Set. Empty set is not always element of every Set.

3. The set $S1 = \{\emptyset\}$ is not a Subset of the set $S2 = \{\{\emptyset\}\}$, although $S1$ is an element of $S2$.

→ Combinations of Set

1. Union ($\cup$)

⇒ $A = \{1, 2\}$
$B = \{1, 3\}$
$A \cup B = \{1, 2, 3\}$

$\{a, b\} \cup \{\ \}$
$= \{a, b\}$

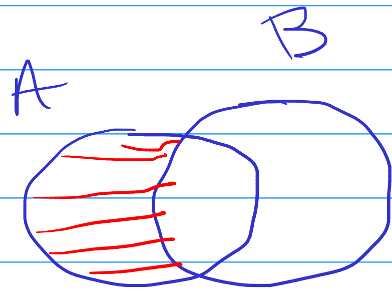
⇒ $\{a, b\} \cup \{\phi\} = \{a, b, \phi\}$

2. Intersection ($\cap$)

⇒ $A \cap B = \{1\}$, $\{a, b\} \cap \{\phi\}$
$= \{\ \}$ OR ϕ

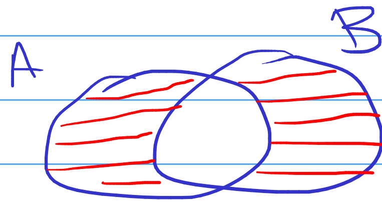
3. Difference

$A = \{a, b, c\}$
$B = \{a\}$
$A - B = \{b, c\}$



4. Symmetric Difference ($\oplus$)

$A \oplus B = (A \cup B) - (A \cap B)$



Power Set

$\mathcal{P}(A)$ is Set of all Possible Subsets of Set A.

- $A = \{1, 2\}$

$\mathcal{P}(A) = \{\underbrace{\{1\}}_{\text{Subset } 1}, \underbrace{\{2\}}_{2}, \underbrace{\{1,2\}}_{3}, \underbrace{\phi}_{4}\}$

~ n elements in a Set, Power Set Contains 2^n elements.

Examples

1. $\phi \subseteq \phi \rightarrow$ ✓
2. $\{a, \phi\} \subseteq \{a, \{a, \phi\}\}$ - F
3. $\{\phi\} \in \{\phi\}$ - F
4. $\{\phi\} \subseteq \phi$ - F